Exact Chord Factorization and Universal Asymptotics for Inscribed Ellipse Polygons

C. Wayne Baker
Independent researcher

GitHub revision v1.3 — October 24, 2026

Abstract

For the ellipse $q(\theta) = (a\cos\theta, b\sin\theta)$, $a \geq b > 0$, let P_N be the perimeter of the equal-parameter inscribed N-gon and let M_N be the composite midpoint approximation to the exact perimeter integral formed from the speed $v(\theta) = (a^2\sin^2\theta + b^2\cos^2\theta)^{1/2}$. We prove the exact factorization $P_N = \operatorname{sinc}(\pi/N)M_N$. For $0 < b < a$, the nearest complex zeros of the squared speed occur at imaginary distance $\rho_* = \operatorname{artanh}(b/a)$, and Fourier aliasing gives $M_N - P = O(e^{-\rho N})$ for every $0 < \rho < \rho_*$. It follows that, for every fixed $r \geq 1$, $(P - P_N)/P = \sum_{k=1}^{r}(-1)^{k+1}\pi^{2k}/((2k+1)!N^{2k}) + O(N^{-2r-2}) + O(e^{-\rho N})$. Thus all algebraic coefficients in the relative polygonal perimeter error are independent of the ellipse axes; shape enters through the exponentially small midpoint remainder and the distance to the nearest complex singularity. Dividing the polygon perimeter by the sinc factor exactly removes the full algebraic chord error, leaving only exponentially convergent midpoint quadrature. Since the speed is also π-periodic, odd numbers of vertices have a doubled exponential aliasing rate. Classical scale-three Richardson cancellation is retained for comparison with geometric N-series constructions.

Keywords: ellipse perimeter; inscribed polygon; midpoint quadrature; periodic quadrature; Fourier aliasing; asymptotic error expansion.

1 Introduction and claim boundary

Polygonal approximation of a smooth closed curve naturally mixes two effects: the geometry of replacing an arc by a chord, and the numerical quadrature implicit in sampling the curve. For an ellipse sampled at equal parameter increments, these two effects separate exactly.

The perimeter of an ellipse has a long history of exact series, inequalities, and closed-form approximations. Ivory's classical work gave a rapidly convergent series for rectification Ivory (1798); later approximation formulas include those associated with Peano and Ramanujan, with modern error analyses and historical discussion available in, for example, Villarino (2006). These approaches concern approximation of the perimeter itself. The present paper instead studies a specific equal-parameter inscribed polygon as a numerical perimeter approximation and identifies an exact factorization linking its geometric chord error to midpoint quadrature of the analytic speed.

The central identity of this paper is

$$P_N = \operatorname{sinc}\left(\frac{\pi}{N}\right) M_N, \tag{1.1}$$

where P_N is the equal-parameter inscribed polygon perimeter and M_N is the composite midpoint approximation to the perimeter integral. The factor $\operatorname{sinc}(\pi/N)$ is universal, whereas all shape dependence is contained in the midpoint sum. Combining this identity with exponential convergence of periodic analytic quadrature shows that every algebraic coefficient in the relative polygonal perimeter error is independent of the ellipse axes.

The analytic quadrature mechanism is classical. Standard references for numerical integration include Davis and Rabinowitz Davis and Rabinowitz (1984); the exponential convergence of periodic trapezoidal rules for analytic functions, together with its Fourier and aliasing interpretation, is treated systematically by Trefethen and Weideman Trefethen and Weideman (2014). Related contour estimates for both trapezoidal and midpoint rules are given by Javed and Trefethen Javed and Trefethen (2014); earlier general contour-based quadrature remainder analysis appears in Donaldson and Elliott (1972).

The contribution is therefore not a new quadrature or extrapolation method. Richardson-type cancellation of algebraic error terms is classical. The purpose here is to isolate the exact geometric factorization (1.1), derive its consequences, and exhibit the resulting universal algebraic coefficients. Historical priority for the bare factorization (1.1) is not asserted here.

For a circle $a = b = R$, the usual regular-polygon identity is

$$P_N = 2NR \sin\frac{\pi}{N} = 2\pi R \operatorname{sinc}\left(\frac{\pi}{N}\right).$$

The ellipse identity below may be viewed as the variable-speed analogue: the exact circumference $2\pi R$ is replaced by a midpoint sampling of the nonconstant ellipse speed.

2 Ellipse geometry and equal-parameter polygons

Fix

$$a \geq b > 0$$

and parametrize the ellipse by

$$q(\theta) = (a\cos\theta, b\sin\theta), \qquad 0 \leq \theta \leq 2\pi. \tag{2.1}$$

Its speed is

$$v(\theta) = \|q'(\theta)\| = \sqrt{a^2\sin^2\theta + b^2\cos^2\theta}. \tag{2.2}$$

The exact perimeter is

$$P = \int_0^{2\pi} v(\theta)\, \mathrm{d}\theta. \tag{2.3}$$

Equivalently,

$$P = 4a\, E\left(1 - \frac{b^2}{a^2}\right),$$

where E denotes the complete elliptic integral of the second kind in parameter notation.

For $N \geq 2$, set

$$h = \frac{2\pi}{N}, \qquad \theta_j = jh, \qquad j = 0, \ldots, N-1,$$

with $\theta_N = 2\pi$. The perimeter of the equal-parameter inscribed polygon is

$$P_N = \sum_{j=0}^{N-1} \|q(\theta_{j+1}) - q(\theta_j)\|. \tag{2.4}$$

Define also the composite midpoint approximation

$$M_N = h \sum_{j=0}^{N-1} v\left(\theta_j + \frac{h}{2}\right). \tag{2.5}$$

3 Exact chord factorization

Theorem 3.1 (Exact chord factorization)**.** *For every $a \geq b > 0$ and every integer $N \geq 2$,*

$$\boxed{P_N = \operatorname{sinc}\left(\frac{\pi}{N}\right) M_N.} \tag{3.1}$$

Proof. For arbitrary real θ and h,

$$\cos(\theta + h) - \cos\theta = -2\sin\left(\theta + \frac{h}{2}\right)\sin\frac{h}{2}$$

and

$$\sin(\theta + h) - \sin\theta = 2\cos\left(\theta + \frac{h}{2}\right)\sin\frac{h}{2}.$$

Hence

$$\begin{aligned}\|q(\theta + h) - q(\theta)\|^2 &= 4\sin^2\frac{h}{2}\left[a^2\sin^2\left(\theta + \frac{h}{2}\right) + b^2\cos^2\left(\theta + \frac{h}{2}\right)\right] \\ &= 4\sin^2\frac{h}{2}\, v\left(\theta + \frac{h}{2}\right)^2.\end{aligned}$$

For $N \geq 2$, $h = 2\pi/N$ and $\sin(h/2) \geq 0$, so

$$\|q(\theta + h) - q(\theta)\| = 2\sin\frac{h}{2}\, v\left(\theta + \frac{h}{2}\right).$$

Summing at $\theta = \theta_j$ gives

$$P_N = 2\sin\frac{\pi}{N}\sum_{j=0}^{N-1} v\left(\theta_j + \frac{\pi}{N}\right).$$

Since

$$M_N = \frac{2\pi}{N}\sum_{j=0}^{N-1} v\left(\theta_j + \frac{\pi}{N}\right),$$

we obtain

$$P_N = \frac{\sin(\pi/N)}{\pi/N}\, M_N.$$

□

Remark 3.2. Theorem 3.1 is exact. No limiting argument, Taylor expansion, or extrapolation method enters its proof.

Remark 3.3 (Literature position)**.** The identity is derived here directly from the standard trigonometric parametrization of the ellipse. Classical ellipse-perimeter formulas and series, such as those discussed in (Ivory, 1798; Villarino, 2006), address the perimeter through elliptic integrals, series, inequalities, or closed-form approximations. No claim of historical priority for the factorization (3.1) is made.

Corollary 3.4 (Exact sinc normalization and exponentially small residual)**.** *For $a \geq b > 0$ and $N \geq 2$ define the sinc-normalized perimeter*

$$\boxed{\widetilde{P}_N := \frac{P_N}{\operatorname{sinc}(\pi/N)} = \frac{\pi P_N}{N \sin(\pi/N)} = M_N.} \tag{3.2}$$

For each fixed noncircular ellipse $0 < b < a$ and each $0 < \rho < \rho_*$,

$$\widetilde{P}_N - P = O(e^{-\rho N}). \tag{3.3}$$

For a circle, $\widetilde{P}_N = P$ identically.

Proof. The equality in (3.2) follows directly from Theorem 3.1. Equation (3.3) then follows from Theorem 5.2. For $a = b$, the speed is constant, so the midpoint rule is exact. □

Remark 3.5 (Computational versus geometric correction)**.** The normalization uses an evaluation of $\sin(\pi/N)$ and a division; it is not asserted to be a straightedge-and-compass operation. It removes the entire sinc-related algebraic error exactly, rather than merely cancelling finitely many powers of N^{-2}. Its advantage is asymptotic for each fixed $b/a > 0$: if $N\rho_*$ is small, the remaining quadrature error may still be significant.

4 Complex singularities of the ellipse speed

Assume first that $0 < b < a$. Consider

$$F(z) = a^2 \sin^2 z + b^2 \cos^2 z.$$

The complex zeros of F determine the obstruction to analytic continuation of the branch of $\sqrt{F}$ that equals v on the real axis.

Theorem 4.1 (Nearest complex singularity distance)**.** *Let $0 < b < a$ and define*

$$\boxed{\rho_* = \operatorname{artanh}\left(\frac{b}{a}\right) = \frac{1}{2}\log\left(\frac{a+b}{a-b}\right).} \tag{4.1}$$

Then $F(z) \neq 0$ throughout

$$|\operatorname{Im} z| < \rho_*,$$

and zeros occur on the boundary at

$$z = n\pi \pm i\rho_*, \qquad n \in \mathbb{Z}.$$

Consequently, the real-positive speed v extends to an analytic branch on every closed substrip

$$|\operatorname{Im} z| \leq \rho, \qquad 0 < \rho < \rho_*.$$

Proof. The equation $F(z) = 0$ is equivalent, whenever $\cos z \neq 0$, to

$$a^2 \tan^2 z + b^2 = 0,$$

hence

$$\tan z = \pm i \frac{b}{a}.$$

Using

$$\tan(iy) = i \tanh y,$$

we obtain the solutions

$$z = n\pi \pm i \operatorname{artanh}\left(\frac{b}{a}\right), \qquad n \in \mathbb{Z}.$$

No zero lies closer to the real axis. Thus F is nonvanishing in the open strip $|\operatorname{Im} z| < \rho_*$. Because this strip is simply connected, the nonvanishing analytic function F admits an analytic square root there, uniquely fixed by requiring agreement with the positive real speed v on the real axis. □

Remark 4.2 (Circle case)**.** If $a = b = R$, then $v(\theta) \equiv R$ is entire and $M_N = P = 2\pi R$ exactly. The formula $\rho_* = \operatorname{artanh}(b/a)$ is then interpreted as the limiting value $+\infty$.

5 Periodic midpoint quadrature

For the ellipse speed, periodicity and continuity on each closed substrip $|\operatorname{Im} z| \leq \rho < \rho_*$ imply the finite bound B_ρ used below.

We now give a self-contained Fourier proof of the exponential midpoint remainder. This avoids requiring a boundary estimate at the singular lines $|\operatorname{Im} z| = \rho_*$.

Lemma 5.1 (Fourier coefficient decay)**.** *Let f be 2π-periodic and analytic in $|\operatorname{Im} z| < \rho_*$. Fix $0 < \rho < \rho_*$, and suppose f is bounded on $|\operatorname{Im} z| \leq \rho$ by B_ρ. If*

$$\widehat{f}_k = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) e^{-ik\theta} \, \mathrm{d}\theta,$$

then

$$|\widehat{f}_k| \leq B_\rho e^{-\rho|k|}.$$

Proof. For $k > 0$, shift the contour from the real axis to $\operatorname{Im} z = -\rho$; periodicity cancels the vertical sides. Then

$$\widehat{f}_k = \frac{1}{2\pi} \int_0^{2\pi} f(\theta - i\rho) e^{-ik(\theta - i\rho)} \, \mathrm{d}\theta,$$

so

$$|\widehat{f}_k| \leq B_\rho e^{-k\rho}.$$

For $k < 0$, shift instead to $\operatorname{Im} z = +\rho$. The case $k = 0$ is immediate. □

Theorem 5.2 (Exponential midpoint remainder)**.** *For every fixed ellipse $0 < b < a$ and every $0 < \rho < \rho_*$,*

$$\boxed{M_N - P = O(e^{-\rho N}).} \tag{5.1}$$

The implied constant may depend on a, b, ρ, but not on N.

Proof. Write the Fourier series

$$v(\theta) = \sum_{k\in\mathbb{Z}} \widehat{v}_k e^{ik\theta}.$$

By Lemma 5.1, the series converges absolutely on the real axis. Since

$$P = 2\pi\widehat{v}_0,$$

and the midpoint nodes are

$$\theta_j^* = \frac{2\pi}{N}\left(j + \frac{1}{2}\right),$$

we have

$$\begin{aligned} M_N &= \frac{2\pi}{N}\sum_{j=0}^{N-1}\sum_{k\in\mathbb{Z}} \widehat{v}_k e^{ik\theta_j^*} \\ &= 2\pi\sum_{m\in\mathbb{Z}} (-1)^m \widehat{v}_{mN}, \end{aligned}$$

because the discrete sum vanishes unless k is a multiple of N, and

$$e^{i(mN)\pi/N} = (-1)^m.$$

Therefore

$$M_N - P = 2\pi\sum_{m\neq 0} (-1)^m \widehat{v}_{mN}.$$

For fixed $0 < \rho < \rho_*$, Lemma 5.1 gives

$$|M_N - P| \le 4\pi B_\rho \sum_{m=1}^{\infty} e^{-\rho m N} = 4\pi B_\rho \frac{e^{-\rho N}}{1 - e^{-\rho N}},$$

which proves (5.1). □

Proposition 5.3 (Parity-refined aliasing of the midpoint rule)**.** *Let $0 < b < a$. For every $0 < \rho < \rho_*$ the sinc-normalized perimeter satisfies*

$$\widetilde{P}_N - P = \begin{cases} O(e^{-\rho N}), & N \text{ even}, \\ O(e^{-2\rho N}), & N \text{ odd}. \end{cases} \tag{5.2}$$

The implied constants may depend on a, b, ρ but not on N.

Proof. The speed has period π, so $v(\theta + \pi) = v(\theta)$ and $\widehat{v}_k = (-1)^k \widehat{v}_k$. Hence $\widehat{v}_k = 0$ for all odd k. The midpoint aliasing identity established in the proof of Theorem 5.2 is

$$M_N - P = 2\pi\sum_{m\neq 0} (-1)^m \widehat{v}_{mN}.$$

If N is even, the first potentially nonzero terms have $m = \pm 1$; if N is odd, m must be even, so the first potentially nonzero terms have $m = \pm 2$. For every $0 < \rho < \rho_*$ the coefficient bound $|\widehat{v}_k| \le B_\rho e^{-\rho|k|}$ implies, respectively,

$$|M_N - P| \le \frac{4\pi B_\rho e^{-\rho N}}{1 - e^{-\rho N}} \quad (N \text{ even}),$$

$$|M_N - P| \le \frac{4\pi B_\rho e^{-2\rho N}}{1 - e^{-2\rho N}} \quad (N \text{ odd}).$$

Use $M_N = \widetilde{P}_N$ to finish. □

Remark 5.4 (Interpretation of the doubled rate)**.** The exponent is doubled at odd N because the π-periodic speed has only even Fourier harmonics. The estimates hold for every strict substrip $\rho < \rho_*$; they do not assert a pure exponential leading asymptotic exactly at the branch-point boundary. Odd N can therefore be especially effective once $N\rho_*$ is large, although no universal advantage at small N is claimed.

6 Universal relative-error expansion

The exact geometry and exponential quadrature estimate now combine.

Theorem 6.1 (Universal finite-order relative-error expansion)**.** *Fix $a \ge b > 0$. If $0 < b < a$, choose any $0 < \rho < \rho_*$; if $a = b$, omit the exponential term below. Then for every fixed integer $r \ge 1$,*

$$\boxed{\frac{P - P_N}{P} = \sum_{k=1}^{r} (-1)^{k+1} \frac{\pi^{2k}}{(2k+1)! N^{2k}} + O(N^{-2r-2}) + O(e^{-\rho N}).} \tag{6.1}$$

In particular, every algebraic coefficient is independent of a and b.

Proof. By Theorem 3.1,

$$\frac{P_N}{P} = \operatorname{sinc}\left(\frac{\pi}{N}\right) \frac{M_N}{P}.$$

For $0 < b < a$, Theorem 5.2 yields

$$\frac{M_N}{P} = 1 + O(e^{-\rho N}).$$

Since $\operatorname{sinc}(\pi/N)$ is bounded,

$$\frac{P_N}{P} = \operatorname{sinc}\left(\frac{\pi}{N}\right) + O(e^{-\rho N}).$$

Now

$$\operatorname{sinc} x = \sum_{k=0}^{r} (-1)^k \frac{x^{2k}}{(2k+1)!} + O(x^{2r+2}) \qquad (x \to 0).$$

Substituting $x = \pi/N$, subtracting from 1, and collecting terms gives (6.1). In the circle case, $M_N = P$ exactly, so the same algebraic expansion follows without an exponential remainder. □

The first few terms are

$$\frac{P - P_N}{P} = \frac{\pi^2}{6N^2} - \frac{\pi^4}{120N^4} + \frac{\pi^6}{5040N^6} - \frac{\pi^8}{362880N^8} + O(N^{-10}) + O(e^{-\rho N}). \tag{6.2}$$

Corollary 6.2 (Universal leading constant)**.** *For every fixed ellipse $a \ge b > 0$,*

$$\boxed{\lim_{N\to\infty} N^2 \frac{P - P_N}{P} = \frac{\pi^2}{6}.} \tag{6.3}$$

Corollary 6.3 (Universal second coefficient)**.** *For every fixed ellipse $a \ge b > 0$,*

$$\lim_{N\to\infty} N^4 \left[\frac{P - P_N}{P} - \frac{\pi^2}{6N^2}\right] = -\frac{\pi^4}{120}.$$

7 Shape dependence and asymptotic onset

Theorems 3.1 and 6.1 separate the two roles played by geometry. The algebraic coefficients arise entirely from the universal chord factor

$$\operatorname{sinc}\left(\frac{\pi}{N}\right),$$

whereas the ellipse shape enters through the midpoint remainder.

For $0 < b < a$,

$$\rho_* = \operatorname{artanh}\left(\frac{b}{a}\right)$$

is the distance to the nearest complex zero of $a^2 \sin^2 z + b^2 \cos^2 z$. As $b/a \downarrow 0$,

$$\rho_* \downarrow 0.$$

Thus highly eccentric ellipses possess a narrower strip of analyticity, and the exponentially small midpoint contribution becomes visible over a longer pre-asymptotic range.

This yields the structural interpretation

universal algebraic coefficients + shape-dependent exponential onset.

No explicit finite-N onset threshold is claimed here.

8 Classical Richardson cancellation as a corollary

We now illustrate how the universal expansion may be used by a standard extrapolation step. This section is deliberately secondary: the cancellation method is classical, while Theorem 6.1 supplies the geometric expansion to which it is applied.

Corollary 8.1 (Scale-3 Richardson cancellation)**.** *Define*

$$\widehat{P}_N = \frac{9P_{3N} - P_N}{8}. \tag{8.1}$$

Then, for every fixed $0 < b < a$ and every $0 < \rho < \rho_*$,

$$\boxed{\frac{P - \widehat{P}_N}{P} = \frac{\pi^4}{1080N^4} - \frac{\pi^6}{40824N^6} + O(N^{-8}) + O(e^{-\rho N}).} \tag{8.2}$$

For a circle the exponential term is absent.

Proof. Write

$$E_N = \frac{P - P_N}{P} = \frac{A}{N^2} + \frac{B}{N^4} + \frac{C}{N^6} + O(N^{-8}) + O(e^{-\rho N}),$$

where

$$A = \frac{\pi^2}{6}, \qquad B = -\frac{\pi^4}{120}, \qquad C = \frac{\pi^6}{5040}.$$

From (8.1),

$$\frac{P - \widehat{P}_N}{P} = \frac{9E_{3N} - E_N}{8}.$$

The N^{-2} term cancels exactly. The remaining coefficients are

$$-\frac{B}{9} = \frac{\pi^4}{1080}$$

and

$$-\frac{10C}{81} = -\frac{\pi^6}{40824}.$$

The exponential contribution remains exponentially small. □

9 Numerical illustrations

The following computations are illustrations of the proved limits, not substitutes for the proofs. The semimajor axis is normalized to $a = 1$, the exact perimeter is evaluated through the complete elliptic integral, and P_N is evaluated from Theorem 3.1.

Table 1: Scaled raw and Richardson-corrected errors for four axis ratios.

b/a	N	$N^2(P - P_N)/P$	$N^4(P - \widehat{P}_N)/P$
0.90	162	1.6449031365344	0.0901927054810
0.50	486	1.6449306301193	0.0901935031057
0.10	1458	1.6449336849891	0.0901935917311
0.01	13122	1.6449340621339	0.0901936026725
Limit	—	$\pi^2/6 = 1.6449340668482$	$\pi^4/1080 = 0.0901936028093$

The increasing values of N used at smaller b/a reflect the shrinking analyticity width ρ_*. The limiting constants themselves do not change with shape.

9.1 Sinc correction and the parity advantage

The sinc normalization uses a single polygon. In contrast, the scale-three Richardson estimate $\widehat{P}_N$ in Corollary 8.1 combines two polygon perimeters. Table 2 illustrates the exponential gain. For the sinc columns, errors are $|\widetilde{P}_N - P|/P$; values are computed with $a = 1$ and the elliptic-integral reference $P = 4E(1 - b^2/a^2)$.

Table 2: Parity effect on the sinc-normalized polygon, $b/a = 0.1$.

N	Parity	Relative error of $\widetilde{P}_N$
95	odd	1.6×10^{-12}
96	even	5.6×10^{-8}
479	odd	4.8×10^{-47}
480	even	9.1×10^{-26}

These results favor sinc normalization when the objective is a numerical perimeter estimate in the resolved quadrature regime. They do not imply superiority to optimized direct algorithms for complete elliptic integrals, and they do not replace the geometric motivation for studying the scale-three extrapolation.

10 Extensions and open questions

The exact factorization in Theorem 3.1 relies on the specific trigonometric parametrization of the ellipse. Several natural questions remain.

First, one may ask which other parametrized closed curves admit an exact decomposition of chord lengths into a universal mesh factor and a sampled local speed or related geometric density. Second, the boundary singularities at $|\operatorname{Im} z| = \rho_*$ have a square-root character; a sharper singularity analysis may yield a more precise asymptotic form than the deliberately conservative $O(e^{-\rho N})$ estimate for $\rho < \rho_*$. Third, the behavior of the approximation in simultaneous limits $N \to \infty$ and $b/a \to 0$ is not covered by the fixed-shape theorem proved here.

No claim is made here concerning arbitrary spherical domains, modular transformations, or a general equivalence between geometric scale cancellation and other acceleration mechanisms.

10.1 Numerical evidence for a sharper midpoint asymptotic

For fixed $0 < b < a$, write $\rho_* = \operatorname{artanh}(b/a)$ and $C(a,b) = 4\sqrt{2\pi ab}$. The nearest zeros of the squared speed are simple and give square-root branch points for v. A Darboux-type analysis therefore suggests the *conjectural* leading laws

$$M_N - P \sim C(a,b) N^{-3/2} e^{-\rho_* N}, \qquad N \text{ even}, \tag{10.1}$$

$$M_N - P \sim -C(a,b)(2N)^{-3/2} e^{-2\rho_* N}, \qquad N \text{ odd}. \tag{10.2}$$

The parity change is consistent with the exact aliasing identity and vanishing of the odd Fourier modes: for odd N, the first nonzero alias occurs at frequency $2N$ with a positive midpoint phase, whereas for even N it occurs at frequency N with a negative midpoint phase. These expressions are *not* asserted as proved asymptotics here.

Table 3: AI-assisted 120-decimal-digit diagnostic for the conjectured even-N leading constant. The scaled quantity is $(M_N - P)N^{3/2}e^{\rho_* N}$, with $a = 1$.

b/a	$N = 20$	$N = 40$	$N = 80$	$N = 160$	$C(1,b)$
0.5	7.43	7.26	7.17	7.13	7.090
0.2	4.88	4.70	4.59	4.54	4.484

A separate AI-assisted calculation reported odd-N scaled values near $-7.25, -7.17, -7.13$ at $N = 21, 41, 81$ with $b/a = 0.5$, compared with $-C(1, 0.5) = -7.090\ldots$. Establishing these estimates with a rigorous branch-point contour analysis, including remainder control, remains open.

11 Reproducibility appendix

The numerical entries in Table 1 can be reproduced without numerical differencing of adjacent ellipse points. For $a = 1$ and a chosen b, evaluate

$$P = 4E(1 - b^2),$$

then compute

$$M_N = \frac{2\pi}{N} \sum_{j=0}^{N-1} \sqrt{\sin^2\left(\left(j + \frac{1}{2}\right)\frac{2\pi}{N}\right) + b^2 \cos^2\left(\left(j + \frac{1}{2}\right)\frac{2\pi}{N}\right)},$$

followed by

$$P_N = \operatorname{sinc}\Big(\frac{\pi}{N}\Big)\, M_N, \qquad \widehat{P}_N = \frac{9P_{3N} - P_N}{8}.$$

High precision should be used when the corrected error is small.

The companion verification procedures can perform these computations using arbitrary-precision arithmetic. The sharp-asymptotic diagnostic uses the explicit midpoint sum with 120-digit working precision.

Data and software availability

No external dataset is required for the theoretical results. The numerical values reported in the paper are reproducible from the explicit formulas stated in the manuscript and the companion verification script retained with the submission materials.

Acknowledgements and disclosure of AI-assisted tools

The author used OpenAI's ChatGPT for editorial drafting, symbolic and numerical cross-checking, code assistance, and manuscript preparation. Anthropic's Claude also supplied an independent AI-assisted critique, including the numerical values in Table 3 and the earlier parity check (Table 2); those displayed values are attributed to that AI-assisted calculation and not represented as a fresh independent author-side run. All mathematical claims, derivations, computations, references, and final wording remain the author's responsibility. No AI system is listed as an author.

References

Donaldson, J. D. & Elliott, D. (1972) A unified approach to quadrature rules with asymptotic estimates of their remainders. *SIAM J. Numer. Anal.*, **9**(4), 573–602. https://doi.org/10.1137/0709051.

Javed, M. & Trefethen, L. N. (2014) A trapezoidal rule error bound unifying the Euler–Maclaurin formula and geometric convergence for periodic functions. *Proc. R. Soc. A*, **470**, 20130571. https://doi.org/10.1098/rspa.2013.0571.

Davis, P. J. & Rabinowitz, P. (1984) *Methods of Numerical Integration*, 2nd edn. Orlando, FL: Academic Press.

Ivory, J. (1798) A new series for the rectification of the ellipsis; together with some observations on the evolution of the formula $(a^2 + b^2 - 2ab\cos\theta)^n$. *Trans. R. Soc. Edinb.*, **4**, 177–190.

Trefethen, L. N. & Weideman, J. A. C. (2014) The exponentially convergent trapezoidal rule. *SIAM Rev.*, **56**, 385–458.

Villarino, M. B. (2006) A note on the accuracy of Ramanujan's approximative formula for the perimeter of an ellipse. *J. Inequal. Pure Appl. Math.*, **7**, Article 21.